Dominance-Support for the Hikita $\star$-product: all lengths, exact up-set support

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WIP commit:	content as of WIP `9330e2c` (the proof was first typeset from `9b9db51`; the source note was last changed in `b5ccc50`, which added Theorem 2 / §8).
Errata (2026-10-01):	§7: 207b grade corrected (Clio's registry has 207b at peer-claimed, her UID 309), not "reviewed by Clio". §6: $n = 6$ check log completed from the logs, the partial $n = 7$ run reported as partial, and the reference to a nonexistent $N = 5$ log fixed. §8.4: N defined. §8: scope of the $n = 6$, $t = 0$ check corrected. No mathematical change.
Errata (2026-10-16):	§7: the 207b inputs are now endorsed as proved by Clio (review 2026-10-15); the "R0" caveat is removed, R0 being Hikita Lemma 3.1. No mathematical change.
Registry:	`proofs/registry/hikita-star-dominance-support.json`, root node, trust: proved

Original header (Day 214 PROVE). Date: 2026-09-30 (deep-work session). Author: Rick.
Status: PROVED for every partition λ, **including the stretch goal**: the support is exactly the full up-set, and $\mathrm{val}_s\, c_{\lambda\mu} = n(\mu)$ (§8). That includes DS_3 (the PROVE.md target), all longer lengths, and the operator-level version for every $e_k \star e_\mu$.
Inputs: 207b $(A_k) + (K_k)$, which give the subset formula (0.1) below, and Lemma 3.3 ($e_k \star 1 = e_k$). Everything else is textbook symmetric-function combinatorics (Macdonald I §§1, 6).
Not used: (TC), ($\bigstar\ell$), the Wick kernel, residues.
Scripts: `scripts/day214/` (exact sympy, all ALL OK).

> **Informal summary.** PROVE.md was bracing for a cross-b cancellation mechanism. There isn't one to find. **Degree bounds don't care about cancellation.** Each subset term of $e_k \star F$ is a rational function whose u-degree under $x_S \to u \cdot x_S$ is at most $\deg_S F + |A \cap S|$, because every kernel factor a_{ij} has degree 0. That is the whole support proof. The (TC) termwise violations were an artefact of expanding in $E(t^i z)E(t^j w)$. The *subset* terms never violate anything. Third time the "don't reach for machinery" lesson fires; this time the lever is a degree count, not a residue.

0. Setup and statement

- $s = q^{-1}$. We work in $\Lambda_m \otimes \mathbb{Q}(s,t)$, writing $K := \mathbb{Q}(s,t)$.
- $a_{ij} := (x_i - tx_j)/(x_i - x_j)$.
- For $A \subseteq [m]$, $\prod_A^{\times} := \prod_{i\in A,\, j\notin A} a_{ij}$ and $X_A := \prod_{a\in A} x_a$.

(0.1) Subset formula. For symmetric F and $1 \le k \le m$,

$$E_k F := e_k \star F = t^{-\binom{k}{2}}\, e_k(Y) \bullet F = \sum_{|A|=k} \prod\nolimits_A^{\times} \cdot X_A \cdot F(X_{A^c}, sX_A). \tag{0.1}$$

Justification.

- 207b (A_k) gives $e_k(Y)F = t^{\binom{k}{2}}\sigma^{(k)}\pi^k F$.
- $\pi^k F = x_1 \cdots x_k\, F(x_{k+1..m}, sx_1..sx_k)$, which is symmetric in the head and in the tail.
- 207b (K_k) gives $\sigma^{(k)}G = \sum_A G^{(A)} \prod_A^{\times}$.
- This is the same formula 207b §4 and (TC) §1 start from.
- Re-checked here: `check_vs_207b.py` compares (0.1) with the 207b closed form symbolically in s,t for $k \le 3$, $r \le 3$.

Because F is symmetric, $F(X_{A^c}, sX_A)$ is simply F with $x_a \mapsto sx_a$ for $a \in A$. The order of the arguments is irrelevant.

Definition. $e_\lambda^{(q,t)} := E_{\lambda_1} E_{\lambda_2} \cdots E_{\lambda_\ell}(1)$. By Lemma 3.3, $E_k(1) = e_k$, so this is $e_{\lambda_1} \star (e_{\lambda_2} \star (\cdots \star e_{\lambda_\ell}))$. The proof below works for the operators applied in **any** order, so neither associativity nor commutativity of $\star$ is used.

Theorem (DS). *For every partition $\lambda \vdash n$, and in Λ_m for every $m \ge n$ (hence in Λ, by §5):*

$$e_\lambda^{(q,t)} = s^{n(\lambda)} e_\lambda + \sum_{\mu \rhd \lambda} c_{\lambda\mu}\, e_\mu, \qquad n(\lambda) = \sum (i-1)\lambda_i,$$

with every $c_{\lambda\mu} \in \mathbb{Q}(t)[s]$ and $c_{\lambda\mu}|_{s=1} = 0$ for $\mu \ne \lambda$. Its parts:

- ***(S) Support:*** *only $\mu \trianglerighteq \lambda$ occur.*
- ***(L) Leading term:*** *the coefficient of e_λ is $q^{-n(\lambda)}$.*
- ***(V) Vanishing:*** *each off-diagonal coefficient is* polynomial *in s, so in particular regular at $s = 1$, and it vanishes there.*

Operator form (Op-DS). *For every $k \ge 1$ and every partition μ:*

$$e_k \star e_\mu \in s^{\sum_i \min(\mu_i, k)}\, e_{\mu \cup k} + \operatorname{span}\{e_\nu : \nu \rhd \mu \cup k\}.$$

This is the "74 operator-level cases" of Day 211, now for all k and μ.

1. From e-dominance to monomial degree bounds

Lemma 1.1. *Let $m \geq n$ and $\rho \vdash n$. Then* $\operatorname{span}\{e_\nu : \nu \trianglerighteq \rho'\} = \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ *in* Λ_m^n.

Proof.

1. By Macdonald I (6.6), $e_\nu = \sum_\kappa M_{\nu\kappa} m_\kappa$, where $M_{\nu\kappa}$ counts 0-1 matrices with row sums ν and column sums κ.
2. If $M_{\nu\kappa} \neq 0$, then $\kappa \trianglelefteq \nu'$. The first j columns of such a matrix hold at most $\sum_i \min(\nu_i, j) = \nu'_1 + \cdots + \nu'_j$ ones.
3. Also $M_{\nu\nu'} = 1$, which is (6.7).
4. By Macdonald I (1.11), $\nu \trianglerighteq \rho' \iff \nu' \trianglelefteq \rho$. So every e_ν on the left lies in the right span.
5. The e_ν ($\nu \vdash n$) are linearly independent in Λ_m, because $m \geq n$. The map $\nu \mapsto \nu'$ is a bijection between the two index sets, so the dimensions agree. □

Degree. For $S \subseteq [m]$ and a nonzero $R \in K(x)$, let $\deg_S R$ be the degree in u of $R|_{x_i \mapsto u x_i \ (i \in S)}$. Here the u-degree of a rational function over $K(x)$ means $\deg \text{num} - \deg \text{den}$.
Properties:

- $\deg_S(R_1 R_2) = \deg_S R_1 + \deg_S R_2$.
- $\deg_S(R_1 + R_2) \leq \max(\deg_S R_1, \deg_S R_2)$.
- For a polynomial P, $\deg_S P$ is the largest S-degree $\sum_{i \in S} \alpha_i$ of a monomial x^α of P. The reason is that the u^d coefficient is the S-degree-d part of P, which is nonzero.
- For symmetric P, $\deg_S P$ depends only on $j = |S|$. Write $D_j(P)$ for it.

Lemma 1.2. *Let $P \in \Lambda_m$ be homogeneous of degree $n = |\rho|$. Then $P \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ if and only if $D_j(P) \leq \rho_1 + \cdots + \rho_j$ for all j.*

Proof. The m_κ have disjoint monomial supports, and $D_j(m_\kappa) = \kappa_1 + \cdots + \kappa_j$. □

2. The degree lemma ⇒ (S)

Lemma 2.1 (degree step)**.** *Let F be a symmetric polynomial and $k \leq m$. For every j,*

$$D_j(E_k F) \leq D_j(F) + \min(k, j).$$

Proof. Fix S with $|S| = j$ and bound each subset term $R_A = \prod_A^\times \cdot X_A \cdot F(X_{A^c}, sX_A)$ of (0.1).

- **The kernel.** $\deg_S a_{ij} = 0$ in all cases:
  - if i and j are both in S, or both out, then a_{ij} is homogeneous of degree 0 in u, or constant;
  - if exactly one is in S, then numerator and denominator both have u-degree 1.
- **The monomial.** $\deg_S X_A = |A \cap S| \leq \min(k, j)$.

- **The shifted F.** $F(X_{A^c}, sX_A)$ is F with some variables multiplied by the constant $s \in K^\times$, and this does not change u-degrees. So its $\deg_S$ equals the maximal S-degree of a monomial of F, which is $D_j(F)$.

Hence $\deg_S R_A \le D_j(F) + \min(k, j)$. The degree of a sum is at most the maximum of the degrees. $E_k F$ is a polynomial by 207b, so its $\deg_S$ is the monomial degree. □

Proof of (S). Induct on the operators applied, in any order. Start from $D_j(1) = 0$. After applying $E_{\lambda_\ell}, \dots, E_{\lambda_1}$,

$$D_j(e_\lambda^{(q,t)}) \le \sum_i \min(\lambda_i, j) = \lambda'_1 + \cdots + \lambda'_j.$$

By Lemma 1.2, $e_\lambda^{(q,t)} \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \lambda'\}$. By Lemma 1.1 this is $\operatorname{span}\{e_\nu : \nu \trianglerighteq \lambda\}$. ■(S)

Op-DS support is the same argument with a single step: starting from $F = e_\mu$ with $D_j = \sum \min(\mu_i, j)$, one gets $D_j \le \sum \min(\mu_i, j) + \min(k, j) =$ the partial sums of $(\mu \cup k)'$.

3. The leading coefficient (L)

Fix integers $w_1 > w_2 > \cdots > w_m > 0$. Substitute $x_i \mapsto u^{w_i} x_i$ and expand in $K(x)((u^{-1}))$. For $\rho = (\rho_1 \ge \cdots \ge \rho_m \ge 0)$, put $W(\rho) := \sum w_i \rho_i$.

Lemma 3.1. *Let x^α be a monomial with $\operatorname{sort}(\alpha) = \kappa \trianglelefteq \rho$. Then $\sum w_i \alpha_i \le W(\rho)$, with equality only if $\alpha = \rho$.*

Proof.

- $\sum w_i \alpha_i \le \sum w_i \kappa_i$ by the rearrangement inequality. Since the w_i are strictly decreasing, equality holds only if α is weakly decreasing, i.e. $\alpha = \kappa$.
- Abel summation gives $\sum w_i \kappa_i = \sum_j (w_j - w_{j+1})(\kappa_1 + \cdots + \kappa_j)$, with $w_{m+1} := 0$. Every weight $w_j - w_{j+1}$ is positive, so this is at most the same expression for ρ. Equality holds only if all the partial sums agree, i.e. $\kappa = \rho$. □

Consequently, if $P \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ has m_ρ-coefficient c, then the $u^{W(\rho)}$-coefficient of $P(u^w x)$ is $c \cdot x^\rho$.

Lemma 3.2 (lead step)**.** *Let $F \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho\}$ with m_ρ-coefficient c_F, and let $k \le m$. Then $E_k F \in \operatorname{span}\{m_\kappa : \kappa \trianglelefteq \rho + 1^k\}$, and its $m_{\rho+1^k}$-coefficient is*

$$s^{\rho_1 + \cdots + \rho_k}\, c_F.$$

Proof.

1. **Membership.** This is Lemma 2.1 together with Lemma 1.2. The partial sums of $\rho + 1^k$ are those of ρ plus $\min(k, j)$.
2. **Expand each factor of R_A in u^{-1}.**
   - **The kernel.** If $w_i > w_j$ $(i < j)$, then $a_{ij} = (1 - t u^{w_j - w_i} x_j/x_i)/(1 - u^{w_j - w_i} x_j/x_i) = 1 + O(u^{-1})$. If $i > j$, then $a_{ij} = t + O(u^{-1})$. So $\prod_A^\times = t^{\operatorname{inv}(A)}(1 + O(u^{-1}))$, where $\operatorname{inv}(A) = \#\{i \in A, j \notin A : i > j\}$.
   - **The monomial.** X_A becomes $u^{\sum_{a \in A} w_a} X_A$.

- **The shifted F.** $F(X_{A^c}, sX_A)$ has the same monomials as F, but x^α is rescaled by $s^{\sum_{a\in A}\alpha_a}$. By Lemma 3.1 it is $u^{W(\rho)}(c_F s^{\sum_{a\in A}\rho_a}x^\rho + O(u^{-1}))$.

3. **Leading orders.** So $R_A = u^{W(\rho)+\sum_A w_a}\big(t^{\mathrm{inv}(A)}c_F s^{\sum_A \rho_a}x^{\rho+1_A} + O(u^{-1})\big)$.

   - $\sum_{a\in A} w_a$ is uniquely maximised at $A = [k]$, where inv $= 0$.
   - Hence the $u^{W(\rho+1^k)}$-coefficient of $E_kF(u^w x)$ is $c_F s^{\rho_1+\cdots+\rho_k}x^{\rho+1^k}$. The other A are of strictly lower order.

4. **Conclude.** Apply the Consequence of Lemma 3.1 to E_kF with the partition $\rho + 1^k$. □

Proof of (L). Apply the operators in any order $p_1, p_2, \ldots, p_\ell$, one after another. When E_{λ_p} is applied to the current $\rho = (\cup_{i \text{ already applied}}\lambda_i)'$, it multiplies the lead by

$$s^{\rho_1+\cdots+\rho_{\lambda_p}} = s^{\sum_{i \text{ applied}}\min(\lambda_i,\lambda_p)}.$$

Over the whole process every unordered pair $\{i, p\}$ contributes $\min(\lambda_i, \lambda_p)$ exactly once. So the total exponent is

$$\sum_{i<p}\min(\lambda_i,\lambda_p) = \sum_{i<p}\lambda_p = \sum_p (p-1)\lambda_p = n(\lambda).$$

The $m_{\lambda'}$-coefficient equals the e_λ-coefficient: in Lemma 1.1's triangularity, the only e_ν ($\nu \trianglerighteq \lambda$) that contains $m_{\lambda'}$ is $\nu = \lambda$, and there with coefficient 1. So $c_{\lambda\lambda} = s^{n(\lambda)} = q^{-n(\lambda)}$. ■(L)

For Op-DS, the lead is $s^{\sum_{i\le k}\mu'_i} = s^{\sum_i \min(\mu_i,k)}$.

4. Vanishing at $s = 1$ (V)

Let $R := \mathbb{Q}(t)[s]$ and $V := \prod_{i<j}(x_i - x_j)$.

1. **Clearing denominators.** Multiplying (0.1) by V gives

$$V \cdot E_kF = \sum_A \varepsilon_A \cdot \prod_{\substack{i<j \\ \text{both in } A \text{ or both out}}}(x_i - x_j) \cdot \prod_{i\in A,\, j\notin A}(x_i - tx_j) \cdot X_A \cdot F(X_{A^c}, sX_A),$$

   where $\varepsilon_A = (-1)^{\mathrm{inv}(A)}$. Call this sum N. If $F \in R[x]$, then $N \in R[x]$.

2. **Coefficients stay in R.** $V \in \mathbb{Z}[x]$ has content 1. By Gauss's lemma over the UFD $R[x]$, $E_kF = N/V$ lies in $R[x]$. By induction, $e_\lambda^{(q,t)} \in R[x]^{S_m}$. Expanding a symmetric polynomial over a ring R in the e-basis keeps the coefficients in R, so every $c_{\lambda\mu} \in \mathbb{Q}(t)[s]$.

3. **Specialising.** Let ev : $R[x] \to \mathbb{Q}(t)[x]$ be the ring map $s \mapsto 1$. Applying it to N kills every s inside $F(X_{A^c}, sX_A)$, so

$$V \cdot \mathrm{ev}(E_kF) = \mathrm{ev}(F) \cdot \sum_A \varepsilon_A(\cdots) = \mathrm{ev}(F) \cdot V \cdot E_k(1) = V \cdot \mathrm{ev}(F) \cdot e_k.$$

   Cancelling V in the domain $\mathbb{Q}(t)[x]$ gives $\mathrm{ev}(E_kF) = e_k \cdot \mathrm{ev}(F)$.

4. **Iterating.** $\mathrm{ev}(e_\lambda^{(q,t)}) = e_\lambda$. Since the e-expansion is unique, $c_{\lambda\lambda}|_{s=1} = 1$ (consistent with $s^{n(\lambda)}$) and $c_{\lambda\mu}|_{s=1} = 0$ for $\mu \neq \lambda$. ■(V)

Remark. At $q = 1$ the $\star$-product is literally the ordinary product. For length 2, (V) was Lemma V ($F_n|_{s=1} = 0$). Lemma V is now a corollary of this structural fact, not a coincidence of the F_n.

5. Stability in m

Let $F \in \Lambda_m$ and set $x_m = 0$ in (0.1). The terms are rational, but their denominators $x_i - x_m$ become $x_i \neq 0$, so the substitution is legitimate.

- **Terms with** $m \in A$ die, because of the factor X_A.
- **Terms with** $m \notin A$ have $a_{im}|_{x_m=0} = 1$, and $F(X_{A^c}, sX_A)|_{x_m=0} = (F|_{x_m=0})(\ldots)$.

So $E_k^{(m)} F|_{x_m=0} = E_k^{(m-1)}(F|_{x_m=0})$. The restriction $\Lambda_m \to \Lambda_{m-1}$ sends $e_\nu \mapsto e_\nu$, and it is injective in degree n when $m - 1 \geq n$. Hence the $c_{\lambda\mu}$ do not depend on $m \geq n$, and DS holds in $\Lambda \otimes \mathbb{Q}(q,t)$. ■

DS is proved for all λ. ■

6. Verification (`scripts/day214/`)

Engine. `ek_subset_engine.py` implements (0.1) exactly, as N/V with the division checked exact, and e-expands the result.
`check_vs_207b.py`**.** Compares (0.1) with the 207b Pieri closed form, *symbolically* in s, t, for $k \leq 3$ and $r \leq 3$ (log `check_vs_207b.log`). This confirms the conventions and normalisation match the proved length-2 theory.
`check_ds_all_lengths.py N`**.** Runs over every $\lambda \vdash n \leq N$, of all lengths, at the exact point $(s,t) = (3/7, -5/11)$, and checks:

- (S);
- (L), i.e. lead $= s^{n(\lambda)}$ exactly;
- (V), by running the engine at $s = 1$ and checking $e_\lambda^{(q,t)} = e_\lambda$;
- order-independence, $E_{\lambda_\ell} \cdots E_{\lambda_1} = E_{\lambda_1} \cdots E_{\lambda_\ell}$;
- "support = full up-set".

Results:

- $n \leq 5$**:** all 18 partitions pass all five checks. Source: the first 18 lines of `check_ds_all_lengths_N6.log`. (There is no separate $N = 5$ log.)
- $n = 6$ **with** `check_ds_all_lengths.py 6`**:** incomplete. The log stops after (6) and $(5,1)$, both of which pass everything. It has no final `ALL OK` line. `check_ds_n.py 6` (log `check_ds_n6.log`; no order-independence check) also stops early, after (6), $(5,1)$, $(4,2)$, all passing.
- $n = 6$**, complete, with** `ds_pointeval.py`**:** this is an independent exact implementation of (0.1), using rational point evaluation in $m = n$ variables. It interpolates the e-coefficients by an exact linear solve, verified at an extra point, at the same $(s,t) = (3/7, -5/11)$, and makes the same five checks. In `ds_pointeval_1_7.log`, all 11 partitions of 6 pass (S), (L), (V), order-independence and full up-set. The same log reproduces all 18 partitions with $n \leq 5$.
- $n = 7$**, partial:** `ds_pointeval_1_7.log` covers 9 of the 15 partitions of 7, from (7) to $(3,2,2)$. `ds_pointeval_7.log` covers 10, from (7) to $(3,2,1,1)$. All of these pass all checks. Neither run finished: there is no final summary line, and $(3,1^4)$, $(2^3,1)$, $(2^2,1^3)$, $(2,1^5)$, (1^7) are unchecked.

How to read the full-up-set results. The $c_{\lambda\mu}$ are polynomials in s with coefficients in $\mathbb{Q}(t)$. So a nonzero value at an exact rational point *proves* $c_{\lambda\mu} \neq 0$. For the partitions checked, "support = full up-set" is therefore proved case by case.
Independent confirmation. Day 211's (TC)-based engine found full up-set support for length 3, $|\lambda| \leq 6$.

7. Gaps, caveats, scope

- **No gap in the argument.**
  - The load-bearing inputs are 207b (A_k) and (K_k): proved, and endorsed as proved by Clio's independent review of 2026-10-15 (Theorem 1 reimplemented end to end, 60/60, $m \leq 4$; §3–§4 endorsed via that check and a line-by-line read). The $\star$-reading rests on Hikita Def. 3.4 and Lemmas 3.1 and 3.3 (arXiv:2503.23597); bijectivity of $q_{(m)}$ is Hikita's Lemma 3.1, so the reading is unconditional (warrant: Clio, review 2026-10-15, Finding 1).
  - Macdonald I (1.11), (6.6) and (6.7) are textbook.
  - Gauss's lemma is textbook.
- **Stretch goal: PROVED in §8.** Every $c_{\lambda\mu}$ with $\mu \trianglerighteq \lambda$ is nonzero, and in fact $\mathrm{val}_s\, c_{\lambda\mu} = n(\mu)$ exactly.
- **Trust.** DS no longer depends on (TC) or $(\bigstar\ell)$. So Clio's review status of those ((TC) reviewed 2026-09-30, UID 306, all-(m,k) grade conditional on 207b §§3–4; $(\bigstar\ell)$ not yet reviewed) does NOT gate DS. DS inherits only the 207b grade.
- **Novelty caution.** The argument is the standard "Macdonald-operator triangularity" degree count (compare Macdonald VI §3 / VI (3.6)–(3.10): D_n^r acts triangularly on m_λ). Since the $\star$-operators have exactly Macdonald's shape — a kernel $\prod^{\times}$ times a shift, here twisted by the X_A factor — it is plausible that Hikita or someone else states DS, or its monomial-triangular form, explicitly. That needs a novelty check before any claim. No browsing was done in this session.
- **What changed about the Day 211 obstruction.** The (TC) termwise dominance violations are real, but they are an artefact of the $E(t^i z)E(t^j w)$ expansion basis. The subset expansion (0.1) is termwise degree-bounded. **Lesson: choose the expansion in which the bound is termwise.**

8. Stretch goal PROVED: the support is exactly the up-set, and $\mathrm{val}_s\, c_{\lambda\mu} = n(\mu)$

Theorem 2. *Let λ be a partition and $\mu \trianglerighteq \lambda$.*

(a) $c_{\lambda\mu} \in \mathbb{Q}[s,t]$, *and* $s^{n(\mu)}$ *divides* $c_{\lambda\mu}$.

(b) $c_{\lambda\mu}(s,0) = s^{n(\mu)}(1 + s \cdot \mathbb{Q}[s])$.

Consequently:

- *every* $c_{\lambda\mu}$ $(\mu \trianglerighteq \lambda)$ *is nonzero, so* $\mathrm{supp}\, e_\lambda^{(q,t)} = \{\mu : \mu \trianglerighteq \lambda\}$ exactly*;*

- *the s-adic valuation of $c_{\lambda\mu}$ is exactly $n(\mu)$;*
- *the lowest coefficient $d_{\lambda\mu}(t) := [s^{n(\mu)}]c_{\lambda\mu} \in \mathbb{Q}[t]$ satisfies $d_{\lambda\mu}(0) = 1$.*

At $s = t = 0$, in the rescaled basis $b_\mu := s^{n(\mu)} e_\mu$, $e_\lambda^{(q,t)}$ is $\sum_{\mu \trianglerighteq \lambda} b_\mu$, ***the zeta function of the dominance order****.*

Notation.

- $\omega(\alpha) := \sum_j \binom{\alpha_j}{2}$ for $\alpha \in \mathbb{Z}^m_{\geq 0}$.
- $B_\mu := \{\beta \in \mathbb{Z}^m_{\geq 0} : \mathrm{sort}(\beta) \trianglelefteq \mu'\}$. By §2, this is the monomial support set of $e_\mu^{(q,t)}$.
- Write $e_\mu^{(q,t)} = \sum_\beta a^\mu_\beta x^\beta$. The coefficient a^μ_β is symmetric in β, because $e_\mu^{(q,t)}$ is a symmetric polynomial.

Key identity. For $A \subseteq [m]$, $\omega(\beta + 1_A) = \omega(\beta) + 1_A \cdot \beta$, since $\binom{b+1}{2} = \binom{b}{2} + b$. This is where s^n comes from.

8.1 Integrality

§4's Gauss-lemma argument works over $\mathbb{Q}[s,t]$, because N_A uses only $(x_i - tx_j)$ and no t^{-1}. So E_k preserves $\mathbb{Q}[s,t][x]^{S_m}$, and hence $c_{\lambda\mu} \in \mathbb{Q}[s,t]$.

Specialising $t = 0$ is a ring map. Moreover $c_{\lambda\mu}(s,0)$ are the e-coefficients of $E^0_{\lambda_1} \cdots E^0_{\lambda_\ell}(1)$, where E^0_k is (0.1) with a_{ij} replaced by $x_i/(x_i - x_j)$.

8.2 Lemma A (valuation bound)

Lemma A (valuation bound). *For t generic, and also at $t = 0$, $\mathrm{val}_s(a^\lambda_\alpha) \geq \omega(\alpha)$ for all α.*

Setup.

- Work over $K = \mathbb{Q}(t)(s^{1/2})$, or $K = \mathbb{Q}(s^{1/2})$ at $t = 0$, with the s-adic valuation v. It is trivial on $\mathbb{Q}(t)$.
- Extend v to $K(x)$ by the Gauss valuation: $v(\sum a_\beta x^\beta) = \min_\beta v(a_\beta)$, and $v(P/Q) = v(P) - v(Q)$. This is a valuation.
- For $c \in (\frac{1}{2}\mathbb{Z})^m$, let σ_c be the K-automorphism $x_j \mapsto s^{-c_j} x_j$.

Claim. $v(\sigma_c e_\lambda^{(q,t)}) \geq \min_{\beta \in B_\lambda}(\omega(\beta) - c \cdot \beta)$ for all c.
Proof of the claim. Induct along E_k: $e_\mu \mapsto e_{\mu \cup k}$. Base case: $1 = e_\varnothing$, with $B_\varnothing = \{0\}$.

For a term R_A of (0.1) applied to $F = e_\mu^{(q,t)}$, the three factors of $\sigma_c R_A$ are:

- **The kernel.** At generic t, $v(\sigma_c a_{ij}) = \min(-c_i, -c_j) - \min(-c_i, -c_j) = 0$. At $t = 0$, $v = -c_i + \max(c_i, c_j) \geq 0$.
- **The monomial.** $\sigma_c X_A = s^{-c \cdot 1_A} X_A$.
- **The shifted F.** $\sigma_c[F(X_{A^c}, sX_A)] = \sigma_{c - 1_A} F$.

Hence

$$v(\sigma_c R_A) \geq -c \cdot 1_A + \min_{\beta \in B_\mu} \big(\omega(\beta) - (c - 1_A) \cdot \beta\big) = \min_{\beta \in B_\mu} \big(\omega(\beta + 1_A) - c \cdot (\beta + 1_A)\big),$$

using the key identity. Now $\beta + 1_A \in B_{\mu \cup k}$: the top-$j$ sums grow by at most $\min(j, k)$, and $\mu' + 1^k = (\mu \cup k)'$. So the bound is $\geq \min_{B_{\mu \cup k}}$. Take the minimum over A. □

Extraction. Put $c := \alpha - \frac{1}{2} \cdot \mathbb{1}$. Then

$$\omega(\beta) - c \cdot \beta = \sum_j \Big[\frac{(\beta_j - \alpha_j)^2}{2} - \frac{\alpha_j^2}{2}\Big],$$

whose *unique* minimiser over all of $\mathbb{Z}^m$ is $\beta = \alpha$. Since $v(\sigma_c P) = \min_\beta(v(a_\beta) - c \cdot \beta)$, we get $v(a_\alpha) - c \cdot \alpha \geq \omega(\alpha) - c \cdot \alpha$. ■(A)

8.3 From monomials to e-coefficients

Recall $n(\nu) = \sum_i (n - S_i(\nu))$, so $\nu \lhd \mu$ implies $n(\nu) > n(\mu)$. Also $n(\mu) = \omega(\mu')$.

Write $c_\nu = s^{n(\nu)} d_\nu$. Suppose $\min_\nu v(d_\nu) = \delta < 0$. Among the ν with $v(d_\nu) = \delta$, pick μ minimal in dominance. Then:

- $a_{\mu'} = \sum_{\nu \trianglelefteq \mu} c_\nu M_{\nu\mu'}$, with $M_{\mu\mu'} = 1$ (Lemma 1.1).
- The $\nu = \mu$ term has valuation exactly $n(\mu) + \delta$.
- Every other term has valuation $> n(\mu) + \delta$: if $\nu \lhd \mu$, then $n(\nu) > n(\mu)$.

So $v(a_{\mu'}) = n(\mu) + \delta < \omega(\mu')$, contradicting Lemma A. Hence every d_ν is s-integral, which proves Theorem 2(a). Reducing mod s gives

$$d_\mu(0) = [s^{\omega(\mu')}]\, a_{\mu'} =: \Lambda_\lambda(\mu').$$

8.4 Lemma B (initial forms at $t = 0$)

Lemma B (initial forms at $t = 0$). *At $t = 0$, $\Lambda^0_\lambda(\alpha) := [s^{\omega(\alpha)}] a^\lambda_\alpha$ equals 1 if $\operatorname{sort}(\alpha) \trianglelefteq \lambda'$, and 0 otherwise.*

Setup. Let $\operatorname{in}_{v_0}(R)$ denote the reduction of $s^{-v_0} R$ in the residue field $\mathbb{Q}(x)$ of the Gauss valuation, defined when $v(R) \geq v_0$. It is additive at a common level, and multiplicative.

Fix α with $\operatorname{sort} \alpha \trianglelefteq (\mu \cup k)'$, put $c = \alpha - \frac{1}{2}$ and $v_0 = \omega(\alpha) - c \cdot \alpha$. By the extraction uniqueness,

$$\operatorname{in}_{v_0}(\sigma_c e^{(q,t)}_{\mu \cup k}) = \Lambda^0_{\mu \cup k}(\alpha)\, x^\alpha.$$

Termwise analysis. Consider the term R_A.

- **Terms that miss the top order.** Suppose $\alpha - 1_A$ is not in B_μ, or has a negative entry. Then $v(\sigma_c R_A) > v_0$ strictly, because the unique minimiser α is not of the form $\beta + 1_A$. So in_{v_0} kills R_A.
- **Terms that survive.** Otherwise, with $\beta_0 = \alpha - 1_A$ and $c - 1_A = \beta_0 - \frac{1}{2}$,

$$\operatorname{in}_{v_0}(\sigma_c R_A) = \operatorname{in}_0(\sigma_c \textstyle\prod_A^\times) \cdot \Lambda^0_\mu(\beta_0) \cdot x^\alpha.$$

The kernel factor at $t = 0$. Here $\mathrm{in}_0(\sigma_c a_{ij})$ equals:

- 1 if $\alpha_i > \alpha_j$;
- 0 if $\alpha_i < \alpha_j$;
- $x_i/(x_i - x_j)$ if $\alpha_i = \alpha_j$.

So only A that are *upper sets* for α survive. Let v be the k-th largest entry of α and $L_v := \{j : \alpha_j = v\}$. The surviving sets are $A = \{\alpha > v\} \sqcup B$, where $B \subseteq L_v$ has size $r := k - \#\{\alpha > v\}$. Every entry of A is ≥ 1, because $\ell(\mathrm{sort}\,\alpha) \geq \ell((\mu \cup k)') \geq k$.
Summing over B. $\mathrm{sort}(\alpha - 1_A) = \mathrm{peel}_k(\mathrm{sort}\,\alpha)$ does not depend on B, and Λ^0_μ is symmetric. So

$$\Lambda^0_{\mu\cup k}(\alpha) = \Lambda^0_\mu(\mathrm{peel}_k\,\alpha) \cdot \sum_{B\subseteq L_v,\ |B|=r} \prod_{i\in B,\ j\in L_v\setminus B} \frac{x_i}{x_i - x_j} = \Lambda^0_\mu(\mathrm{peel}_k\,\alpha).$$

The last sum is 1. It is symmetric (S_{L_v} permutes the terms), so its product with the Vandermonde is antisymmetric, and the sum is a polynomial. It is homogeneous of degree 0, hence a constant. The constant is its u-leading coefficient under the weights $w_1 > w_2 > \cdots$ of §3, where only $B =$ the first r indices contributes, with value 1. The t-version, with a_{ij} in place of $x_i/(x_i - x_j)$, is $\Sigma = \binom{N}{r}_t$, where $N := |L_v|$ is the size of the level set. `check_stretch.py` checks this for $N \leq 4$. It checks the $t = 0$ identity for $N \leq 5$.

Here $\mathrm{peel}_k(\kappa)$ means: subtract 1 from the k largest parts of κ.

Peel Lemma. *Let $\kappa \trianglelefteq \lambda$ with $|\kappa| = |\lambda|$ and $k \leq \ell(\lambda)$. Then $\mathrm{peel}_k\,\kappa \trianglelefteq \mathrm{peel}_k\,\lambda$.*

Proof.

1. **Conjugate formula.** Counting the parts $\geq c$ after the peel gives

   $$(\mathrm{peel}_k\,\kappa)'_c = \kappa'_c - \min(k, \kappa'_c) + \min(k, \kappa'_{c+1}).$$

   (This is checked for $n \leq 12$ in `check_stretch.py`.)

2. **Partial sums.** Telescoping, and using $\kappa'_1 = \ell(\kappa) \geq \ell(\lambda) \geq k$,

   $$S_C((\mathrm{peel}_k\,\kappa)') = S_C(\kappa') - k + \min(k, \kappa'_{C+1}).$$

   The same holds for λ.

3. **Compare.** We have $\kappa' \trianglerighteq \lambda'$ (Macdonald I (1.11)). Put $D_C := S_C(\kappa') - S_C(\lambda') \geq 0$ and $m(x) := \min(k, x)$.

   - If $m(\kappa'_{C+1}) \geq m(\lambda'_{C+1})$, we are done.
   - Otherwise $\kappa'_{C+1} < k$ and $\kappa'_{C+1} < \lambda'_{C+1}$. Then $D_C \geq D_{C+1} + \lambda'_{C+1} - \kappa'_{C+1} \geq m(\lambda'_{C+1}) - m(\kappa'_{C+1})$.

4. So $(\mathrm{peel}_k\,\kappa)' \trianglerighteq (\mathrm{peel}_k\,\lambda)'$, which is $\mathrm{peel}_k\,\kappa \trianglelefteq \mathrm{peel}_k\,\lambda$. □

(A brute-force check confirms it for all 10,788 triples with $n \leq 11$, in `peel_monotone.py`. This is the greedy step of Ryser's algorithm for the Gale–Ryser theorem.)

Proof of Lemma B. Induct along E_k. The base case is $\Lambda^0_\varnothing(0) = 1$. For $\mathrm{sort}\,\alpha \trianglelefteq (\mu \cup k)'$, the Peel Lemma with $\lambda := (\mu \cup k)'$ gives:

- $\mathrm{peel}_k(\mathrm{sort}\,\alpha) \trianglelefteq \mathrm{peel}_k((\mu \cup k)')$;
- $\mathrm{peel}_k((\mu \cup k)') = \mu'$, because $(\mu \cup k)' = \mu' + 1^k$ and its top k parts sit in positions $1..k$.

So $\Lambda^0_{\mu\cup k}(\alpha) = 1$. Outside $B_{\mu\cup k}$, the value is 0 by (S). ■(B)

Proof of Theorem 2. $d_{\lambda\mu}(0)|_{t=0} = \Lambda^0_\lambda(\mu') = [\mu' \trianglelefteq \lambda'] = [\mu \trianglerighteq \lambda] = 1$. Since $c_{\lambda\mu}/s^{n(\mu)} \in \mathbb{Q}[s,t]$ (by §8.3, together with $s^{n(\mu)} \mid c$ in $\mathbb{Q}(t)[s] \cap \mathbb{Q}[s,t]$), its value at $s = t = 0$ is 1. ■

Checks.

- `s_valuation.py` ($n \le 5$), `check_stretch.py` ($n \le 5$) and `check_stretch_n6.py` ($n = 6$, $t = 0$; its log `check_stretch_n6_t0.log` covers 10 of the 11 partitions of 6, all except (1^6)): $\mathrm{val}_s\, c_{\lambda\mu} = n(\mu)$ for every $\mu \trianglerighteq \lambda$, with lowest coefficient exactly 1 at $t = 0$. At $t = -5/11$ the valuation is again $n(\mu)$.
- The $d_{\lambda\mu}(t)$ are genuinely t-dependent. Example: $d_{(1111),(211)} = 1 + 3t$, a value consistent with the data at $t = -5/11$.
- *Remark (computed for $n \le 5$ in `t1_count.py`; the $t = 1$ case follows from the same* in_0 *argument, but it is not written out).* At $t = 1$ the recursion becomes $\Lambda_{\mu\cup k}(\alpha) = \sum_{A\subseteq \mathrm{supp}\,\alpha} \Lambda_\mu(\alpha - 1_A)$. This counts 0-1 matrices, so $d_{\lambda\mu}(1) = M_{\lambda\mu'}$. The case $1+3t$ at $t = 1$ gives $4 = M_{(1111),(31)}$. At general t, the in_0 of a_{ij} for $\alpha_i < \alpha_j$ is t, and the level-set sums are t-binomials. So $d_{\lambda\mu}(t)$ is a t-count of 0-1 matrices with row sums λ and column sums μ'. This is not written out here.